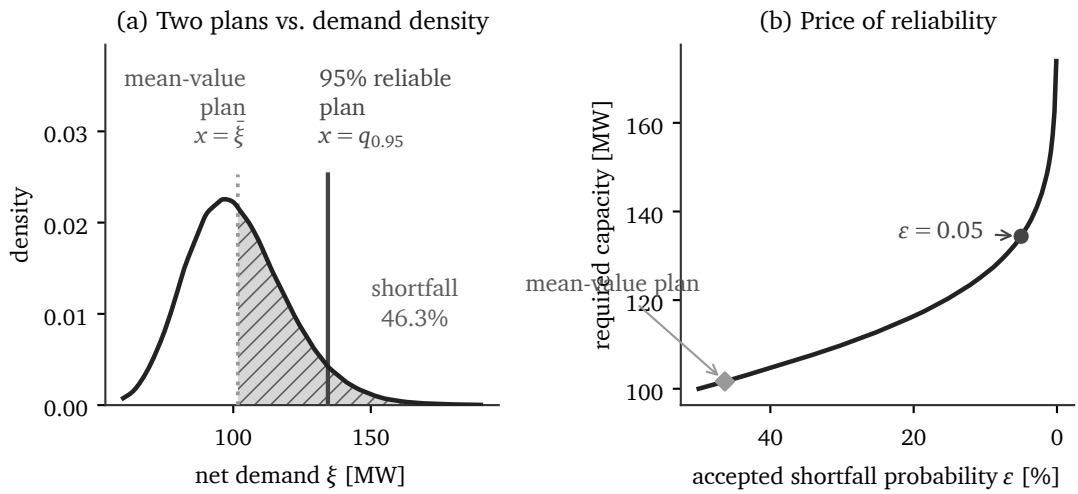
(a) Two plans vs. demand density
mean-value plan
$x = \bar{\xi}$
95% reliable plan
$x = q_{0.95}$
shortfall
46.3%
density
0.03
0.02
0.01
0.00
100
150
net demand ξ [MW]
(b) Price of reliability
$\varepsilon = 0.05$
mean-value plan
required capacity [MW]
160
140
120
100
40
20
0
accepted shortfall probability ε [%]